

non-rel. Schrödinger equation
or relativistic Dirac equation

LDA or GGA
or hybrids

physical orbitals or not
mesh density and basis set

The diagram illustrates the Kohn-Sham equation and its components with the following labels and arrows:

- non-rel. Schrödinger equation or relativistic Dirac equation**: Points to the kinetic energy term.
- LDA or GGA or hybrids**: Points to the exchange-correlation potential term.
- physical orbitals or not**: Points to the orbital term $\varphi_i(\vec{r})$ on the left.
- mesh density and basis set**: Points to the orbital term $\varphi_i(\vec{r})$ on the right.
- pseudopotential (ultrasoft/PAW/norm-conserving) or all-electron**: Points to the external potential term $v_{\text{ext}}(\vec{r})$.
- Hartree potential from solving Poisson eq. or integrating charge density**: Points to the Hartree potential term $v_H(\vec{r})$.
- view EVs as mere Lagrange multipliers or band structure approx**: Points to the energy eigenvalue E_i .

The equation is presented as:

$$\left(-\frac{\hbar^2}{2m} \vec{\nabla}_{\vec{r}}^2 + v_{\text{ext}}(\vec{r}) + v_H(\vec{r}) + v_{\text{xc}} \right) \varphi_i(\vec{r}) = E_i \varphi_i(\vec{r})$$